

Axion App Deployment – Monolithic Architecture

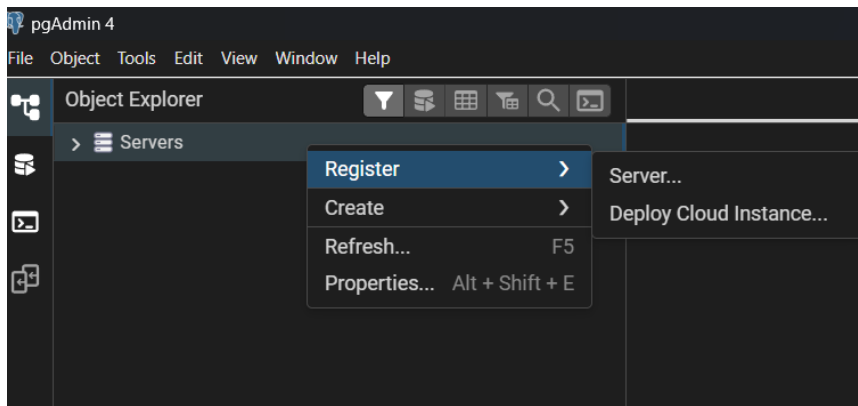
Create Infra using Terraform Code

Repo link - <https://github.com/shyam-codeops/axion-app-monolithic-Infra.git>

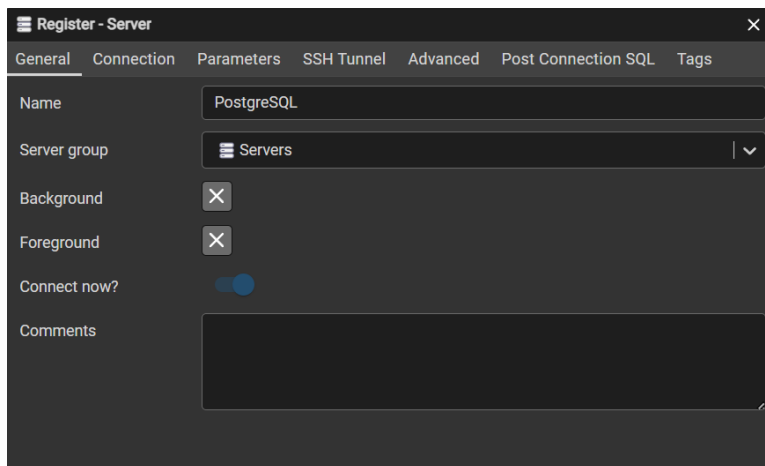
Install pgadmin4

Open pgadmin4

Right click on Servers >> Register



Type Name(Any)



Go to Connection Tab

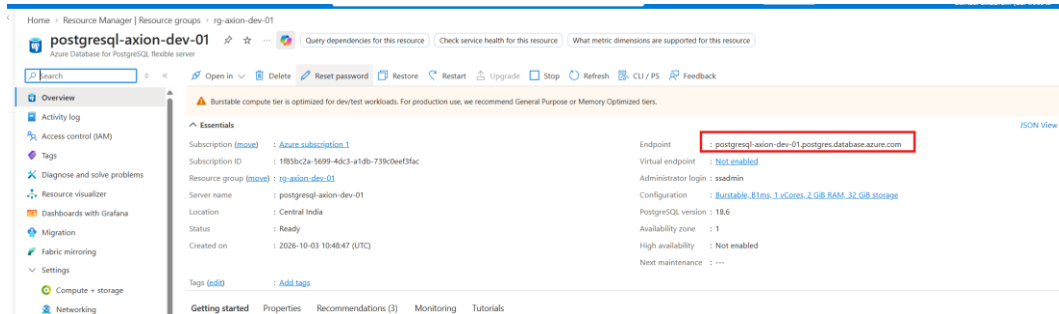
Open azure portal >> Copy Endpoint of Database

Paste that in Hostname/address in pgadmin4

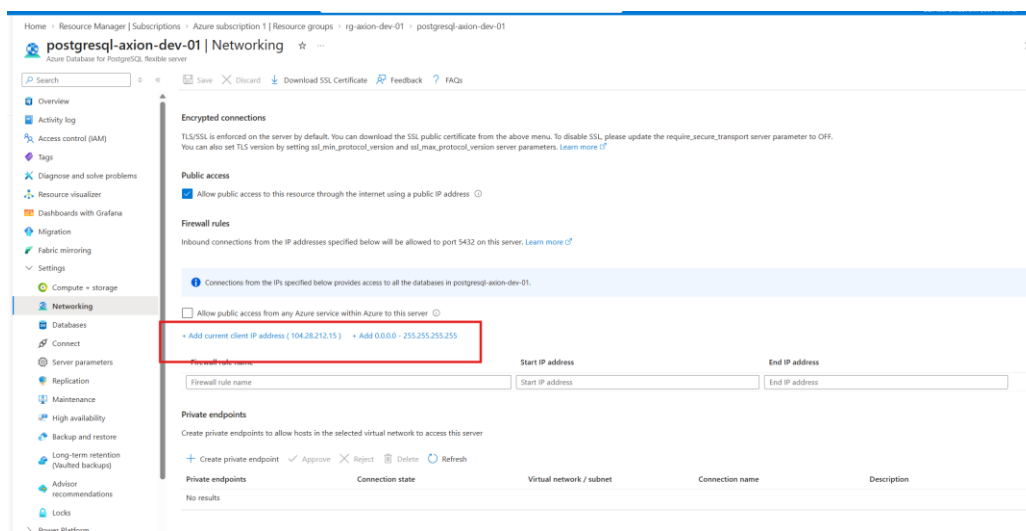
postgresql-axion-dev-01.postgres.database.azure.com

Type username and Password (Mentioned in Terraform Code in Environment/Dev tfvars file)

Do not click Save Yet>> Need to allow Your PC IP on Database



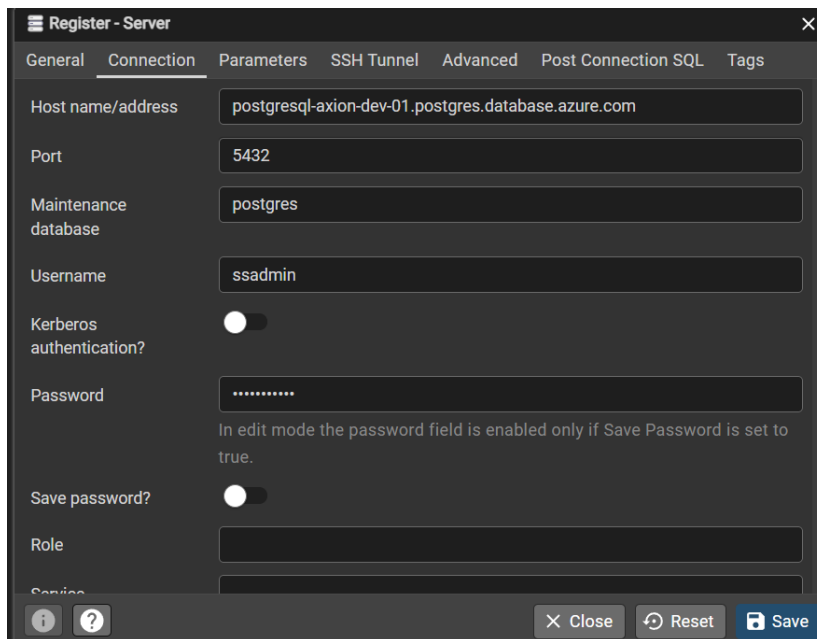
Add Current Client IP(Your PC IP) to Firewall Allow Rule



Click Add and then click Save

Once IP is added.

Click Save in pgadmin4 window

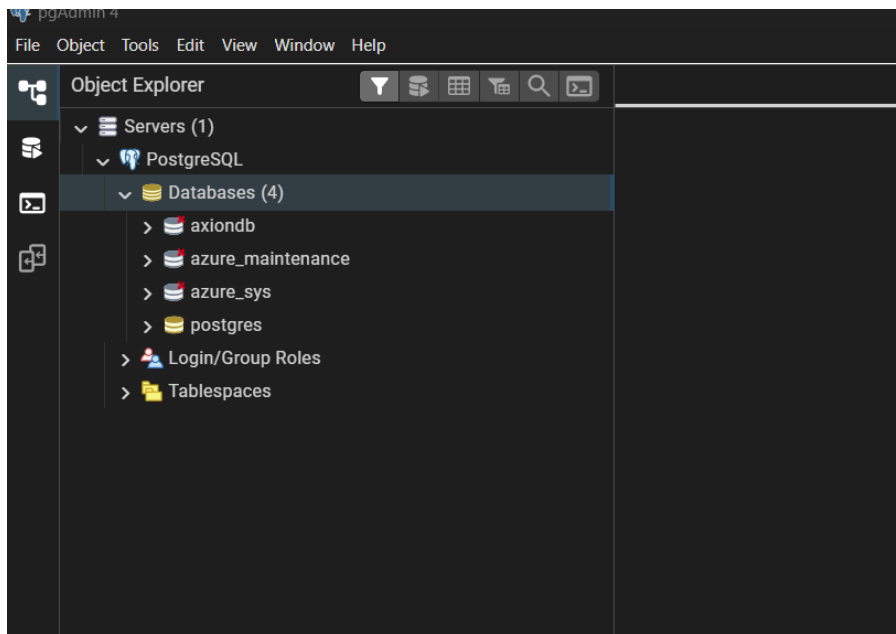


The image shows the 'Register - Server' dialog box in pgAdmin 4, specifically the 'Connection' tab. The fields are filled with the following values:

- Host name/address: postgresql-axion-dev-01.postgres.database.azure.com
- Port: 5432
- Maintenance database: postgres
- Username: ssadmin
- Kerberos authentication?: ☐
- Password: [masked with dots]
- Save password?: ☐
- Role: [empty]
- Service: [empty]

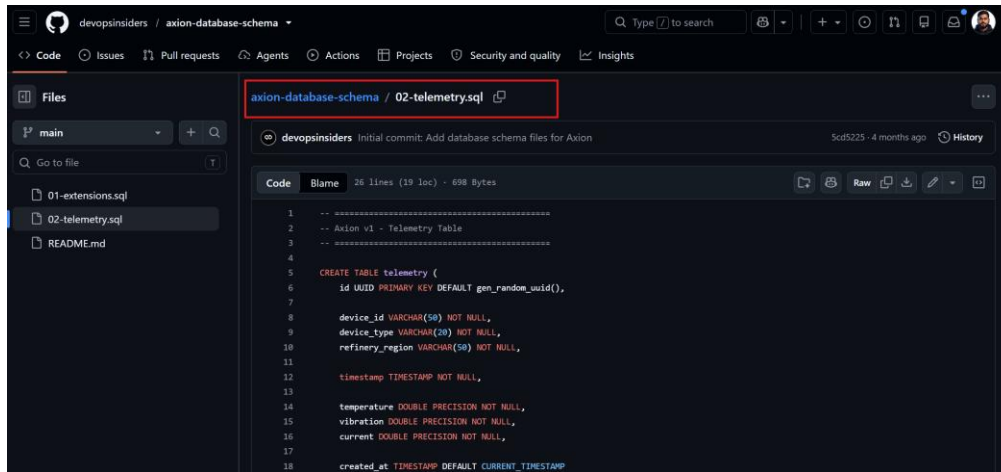
At the bottom right, there are three buttons: 'Close', 'Reset', and 'Save'.

PostgreSQL Database will start showing up



Open <<https://github.com/devopsinsiders/axion-database-schema.git>>

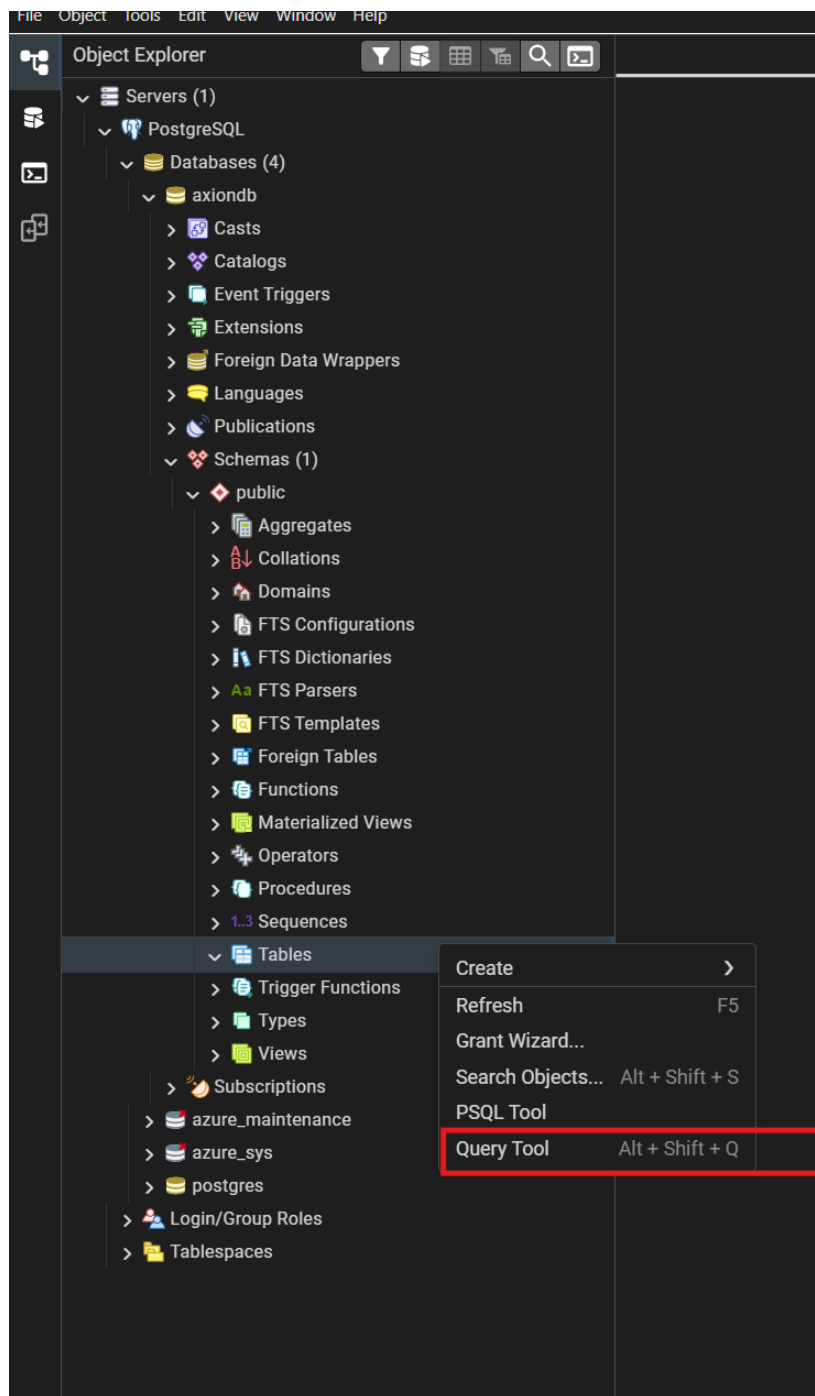
Click on 02-telemetry.sql



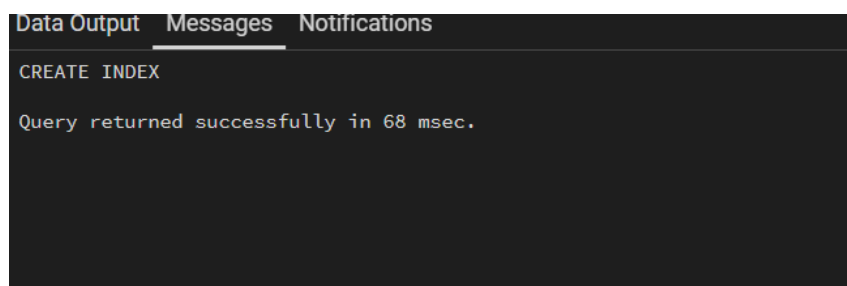
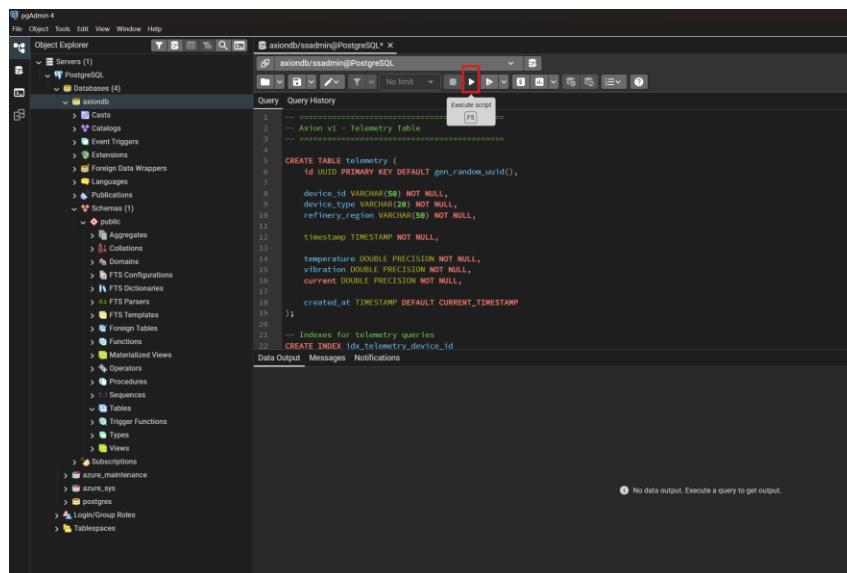
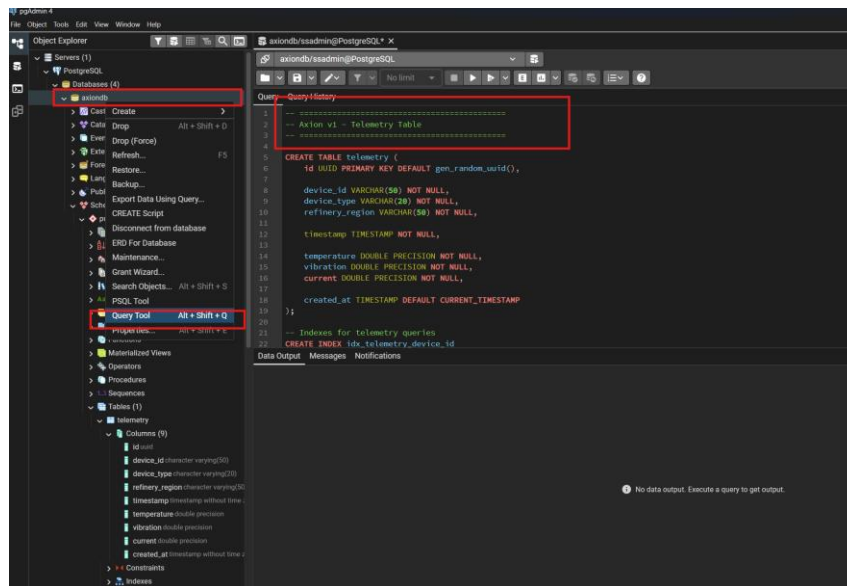
The screenshot shows the GitHub interface for the repository 'devopsinsiders / axion-database-schema'. The file '02-telemetry.sql' is highlighted in the file explorer on the left. The main content area displays the SQL code for creating a 'telemetry' table. The code is as follows:

```
1 -- =====
2 -- Axion v1 - Telemetry Table
3 -- =====
4
5 CREATE TABLE telemetry (
6   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
7
8   device_id VARCHAR(50) NOT NULL,
9   device_type VARCHAR(20) NOT NULL,
10  refinery_region VARCHAR(50) NOT NULL,
11
12  timestamp TIMESTAMP NOT NULL,
13
14  temperature DOUBLE PRECISION NOT NULL,
15  vibration DOUBLE PRECISION NOT NULL,
16  current DOUBLE PRECISION NOT NULL,
17
18  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

In pgadmin4, Right click on axiondb > Tables >> Query Tool

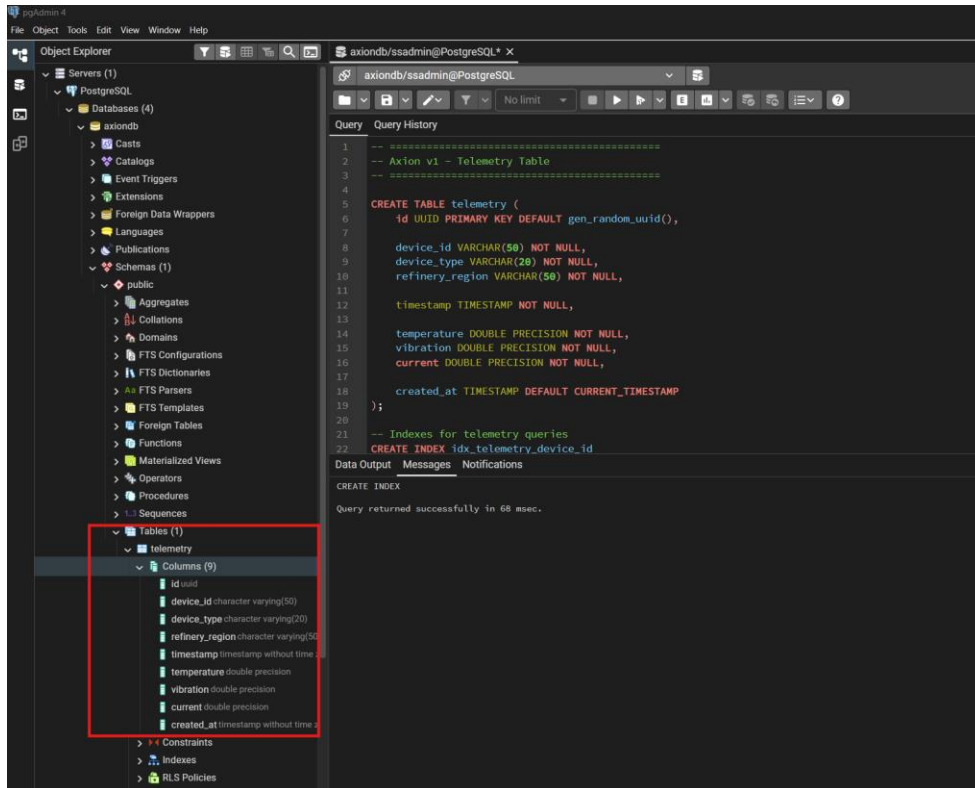


Copy Query from GitHub and Paste in Query Tab and then click on Execute script



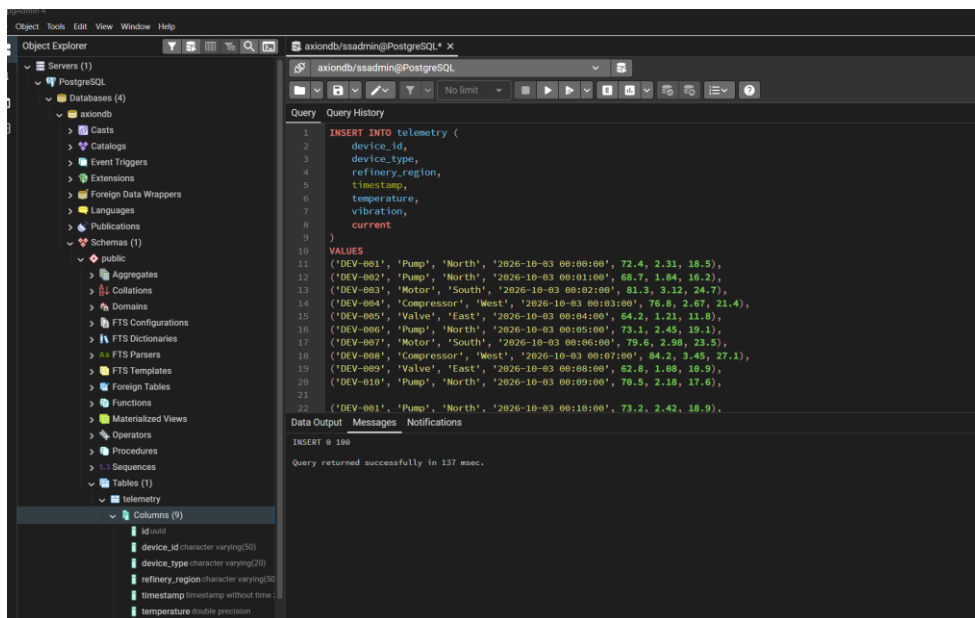
Refresh Database in pgadmin4

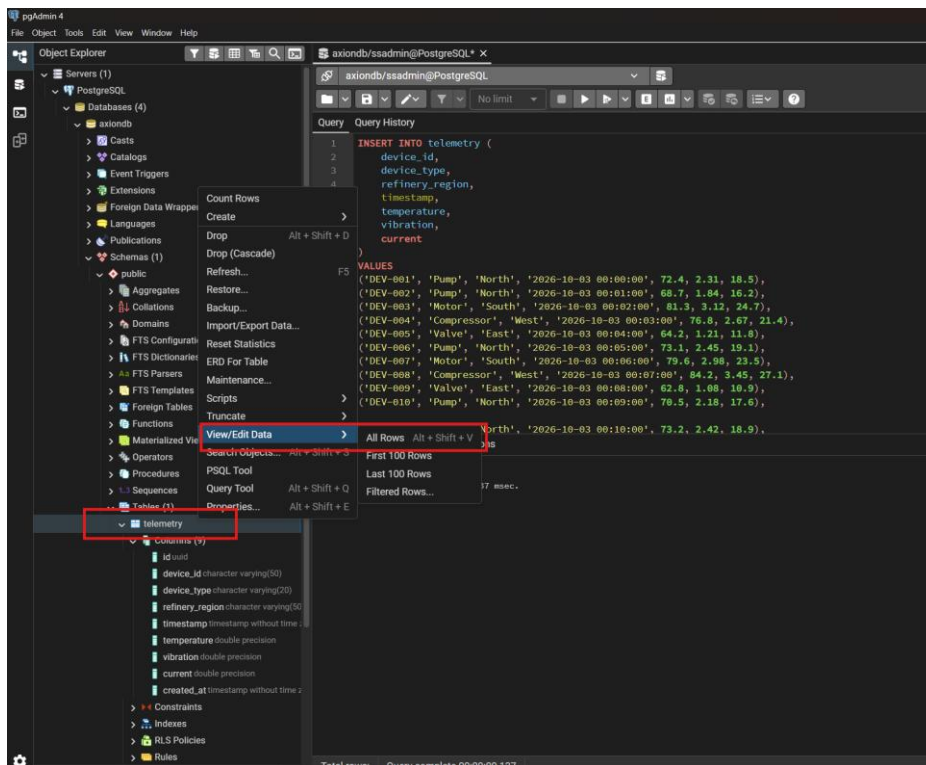
Telemetry Table and Columns got created



Open Dummydata.md

Paste Dummy Data in Query Tool and Run





Dummy Data will show up

id	device_id	device_type	refinery_region	timestamp	temperature	vibration	current	created_at
1	DEV-001	Pump	North	2026-10-03 00:00:00	72.4	2.31	18.5	
2	DEV-002	Pump	North	2026-10-03 00:01:00	68.7	1.84	16.2	
3	DEV-003	Motor	South	2026-10-03 00:02:00	81.3	3.12	24.7	
4	DEV-004	Compressor	West	2026-10-03 00:03:00	76.8	2.67	21.4	
5	DEV-005	Valve	East	2026-10-03 00:04:00	64.2	1.21	11.8	
6	DEV-006	Pump	North	2026-10-03 00:05:00	73.1	2.45	19.1	
7	DEV-007	Motor	South	2026-10-03 00:06:00	79.6	2.98	23.5	
8	DEV-008	Compressor	West	2026-10-03 00:07:00	84.2	3.45	27.1	
9	DEV-009	Valve	East	2026-10-03 00:08:00	62.8	1.08	10.9	
10	DEV-010	Pump	North	2026-10-03 00:09:00	70.5	2.18	17.6	
11	DEV-011	Pump	North	2026-10-03 00:10:00	73.2	2.42	18.9	
12	DEV-012	Motor	South	2026-10-03 00:11:00	82.1	3.25	25.3	
13	DEV-013	Compressor	West	2026-10-03 00:12:00	77.9	2.81	22.8	
14	DEV-014	Valve	East	2026-10-03 00:13:00	65.3	1.35	12.4	
15	DEV-015	Pump	North	2026-10-03 00:14:00	74.6	2.59	20.1	
16	DEV-016	Motor	South	2026-10-03 00:15:00	80.5	3.08	24.2	
17	DEV-017	Compressor	West	2026-10-03 00:16:00	78.2	2.74	23.1	
18	DEV-018	Valve	East	2026-10-03 00:17:00	63.7	1.15	11.3	
19	DEV-019	Pump	North	2026-10-03 00:18:00	72.9	2.38	18.7	
20	DEV-020	Motor	South	2026-10-03 00:19:00	81.7	3.18	24.9	
21	DEV-021	Compressor	West	2026-10-03 00:20:00	76.5	2.65	21.7	
22	DEV-022	Valve	East	2026-10-03 00:21:00	64.8	1.28	12.1	
23	DEV-023	Pump	North	2026-10-03 00:22:00	73.5	2.48	19.3	
24	DEV-024	Motor	South	2026-10-03 00:23:00	79.8	2.95	23.8	
25	DEV-025	Compressor	West	2026-10-03 00:24:00	84.5	3.48	27.4	
26	DEV-026	Valve	East	2026-10-03 00:25:00	62.9	1.09	11.0	
27	DEV-027	Pump	North	2026-10-03 00:26:00	70.7	2.19	17.8	
28	DEV-028	Motor	South	2026-10-03 00:27:00	81.9	3.15	24.6	
29	DEV-029	Compressor	West	2026-10-03 00:28:00	77.4	2.78	22.5	
30	DEV-030	Valve	East	2026-10-03 00:29:00	64.5	1.32	12.2	

Backend

Login to Backend VM

```
ssh username@publicip
```

```
git clone https://github.com/devopsinsiders/axion-telemetry-query-service.git
```

```
cd axion-telemetry-query-service
```

```
sudo apt update
```

```
sudo apt install -y python3.12-venv
```

```
python3 -m venv venv
```

```
source venv/bin/activate # On Windows use: venv\Scripts\activate
```

```
pip install -r requirements.txt
```

```
nano config.py
```

Update config.py

If there is @ in the password use %40

Like if Password is Nested@1234, Update it as Nested%401234 in config.py

Update axion user to ssadmin

Update password

Update localhost to database endpoint url

Update database name to axiondb

Before

```

GNU nano 7.2
"""
Axion Telemetry Query Service - Configuration
"""

import os
from dataclasses import dataclass

@dataclass
class Settings:
    # PostgreSQL connection string
    # Format: postgresql://<user>:<password>@<host>:<port>/<database>
    DATABASE_URL: str = os.getenv(
        "DATABASE_URL",
        "postgresql://axion_user:P%40ssw01rd%40123@localhost:5432/axion_db",
    )
    # Allows configuring a different port, e.g., if we run multiple services
    PORT: int = int(os.getenv("PORT", "8000"))

settings = Settings()

```

After

```

GNU nano 7.2 config.py *
"""
Axion Telemetry Query Service - Configuration
"""

import os
from dataclasses import dataclass

@dataclass
class Settings:
    # PostgreSQL connection string
    # Format: postgresql://<user>:<password>@<host>:<port>/<database>
    DATABASE_URL: str = os.getenv(
        "DATABASE_URL",
        "postgresql://ssadmin:Nested%401234@postgresql-axion-dev-01.postgres.database.azure.com:5432/axion_db",
    )
    # Allows configuring a different port, e.g., if we run multiple services
    PORT: int = int(os.getenv("PORT", "8000"))

settings = Settings()

```

Save using Ctrl + S and Exit using CTRL+X

Check using cat.config.py

```
(venv) ssadmin@backend-vm-axion-dev-01:~/axion-telemetry-query-service$ cat config.py
"""
Axion Telemetry Query Service - Configuration
"""

import os
from dataclasses import dataclass

@dataclass
class Settings:
    # PostgreSQL connection string
    # Format: postgresql://<user>:<password>@<host>:<port>/<database>
    DATABASE_URL: str = os.getenv(
        "DATABASE_URL",
        "postgresql://ssadmin:Nested%401234@postgresql-axion-dev-01.postgres.database.azure.com:5432/axiondb",
    )
    # Allows configuring a different port, e.g., if we run multiple services
    PORT: int = int(os.getenv("PORT", "8000"))

settings = Settings()
(venv) ssadmin@backend-vm-axion-dev-01:~/axion-telemetry-query-service$
```

Allow Backend VM IP in PostgreSQL Database Networking Firewall Rule

The screenshot shows the 'Networking' settings for the PostgreSQL database instance 'postgresql-axion-dev-01'. The 'Firewall rules' section is expanded, showing a table of rules. A red box highlights the 'BackendVM' rule, which has a start IP address of '20.219.192.17' and an end IP address of '20.219.192.17'. The rule is set to allow access to port 5432.

Firewall rule name	Start IP address	End IP address
ClientMachines_2020-10-2-23-35-41	104.28.212.15	104.28.212.15
BackendVM	20.219.192.17	20.219.192.17

Allow 8000 Port on Backend VM

The screenshot shows the 'Network settings' for the Backend VM 'backend-vm-axion-dev-01'. The 'Network security group' 'nsg-backend-axion-dev-01' is expanded, showing a table of rules. A red box highlights the 'PostgreSQL-Allow' rule, which has a priority of 110, a name of 'PostgreSQL-Allow', a port of 8000, and an action of 'Allow'. The rule is set to allow access to port 8000.

Priority	Name	Port	Protocol	Source	Destination	Action
100	AllowSSH	22	TCP	Any	Any	Allow
110	PostgreSQL-Allow	8000	Any	104.28.212.15	Any	Allow

Run

```
uvicorn main:app --host 0.0.0.0 --port 8000
```

It should start running.

```
(venv) ssadmin@backend-vm-axion-dev-01:~/axion-telemetry-query-service$ uvicorn main:app --host 0.0.0.0 --port 8000
INFO: Started server process [7619]
INFO: Waiting for application startup.
Connecting to database...
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
█
```

Copy backend Public IP and port as 8000

<http://13.71.1.67:8000>

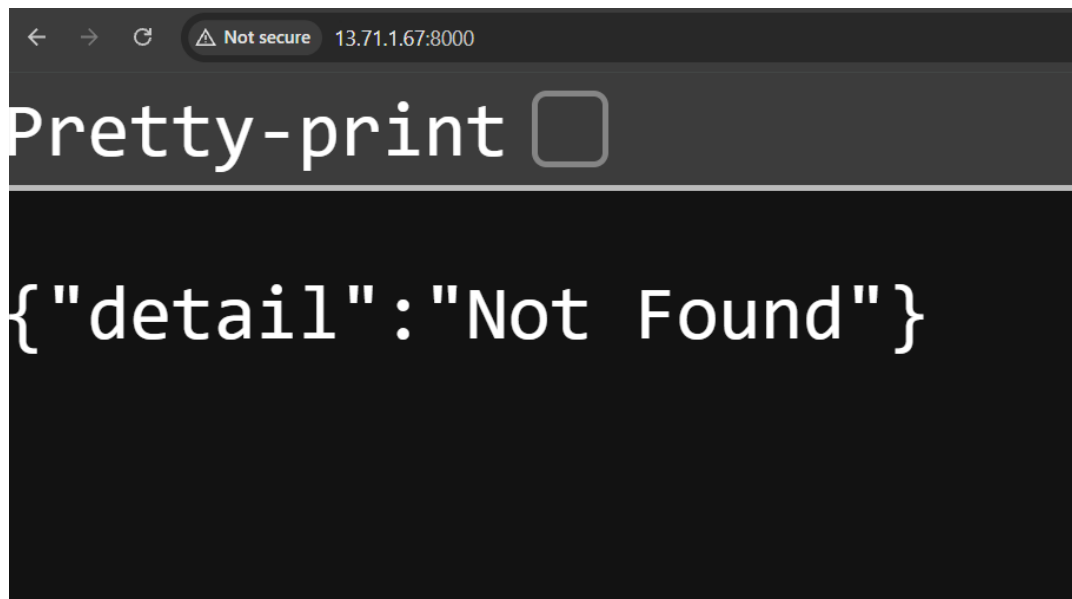
Accessing the API

Once the service is running, you can access it from any browser or HTTP client using your VM's public IP address:

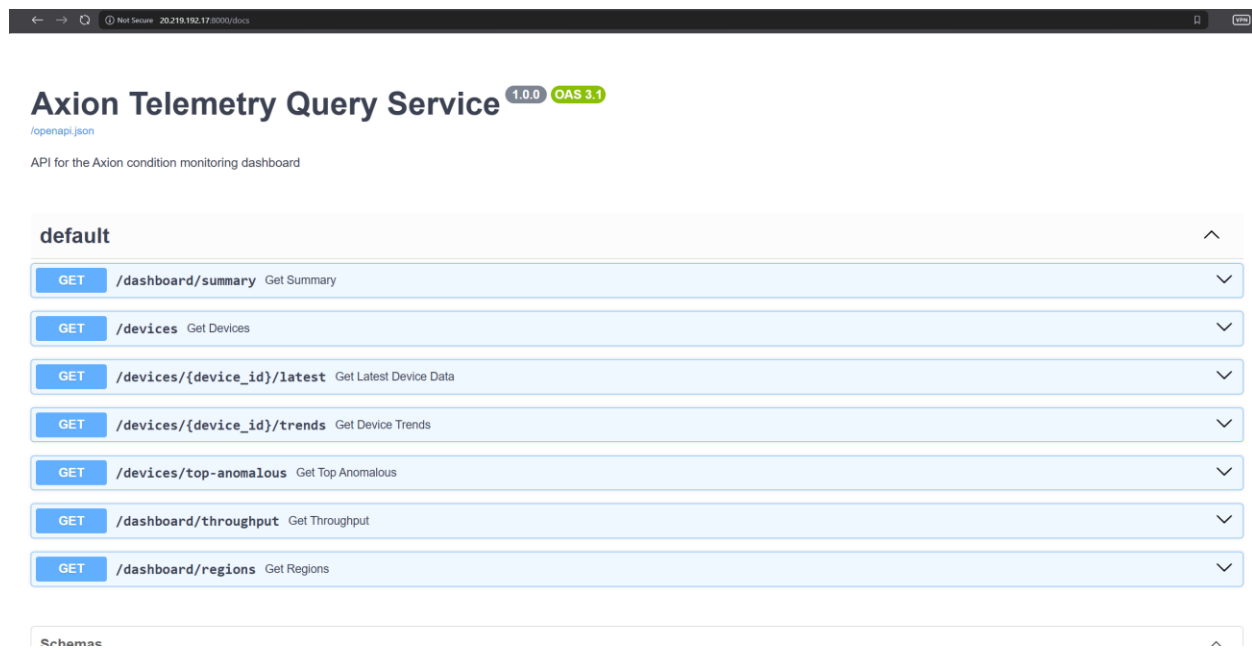
- **API Root:** http://<YOUR_VM_PUBLIC_IP>:8000/
- **Swagger UI Documentation:** http://<YOUR_VM_PUBLIC_IP>:8000/docs
- **Redoc Documentation:** http://<YOUR_VM_PUBLIC_IP>:8000/redoc

Replace <YOUR_VM_PUBLIC_IP> with the actual public IP address of your virtual machine.

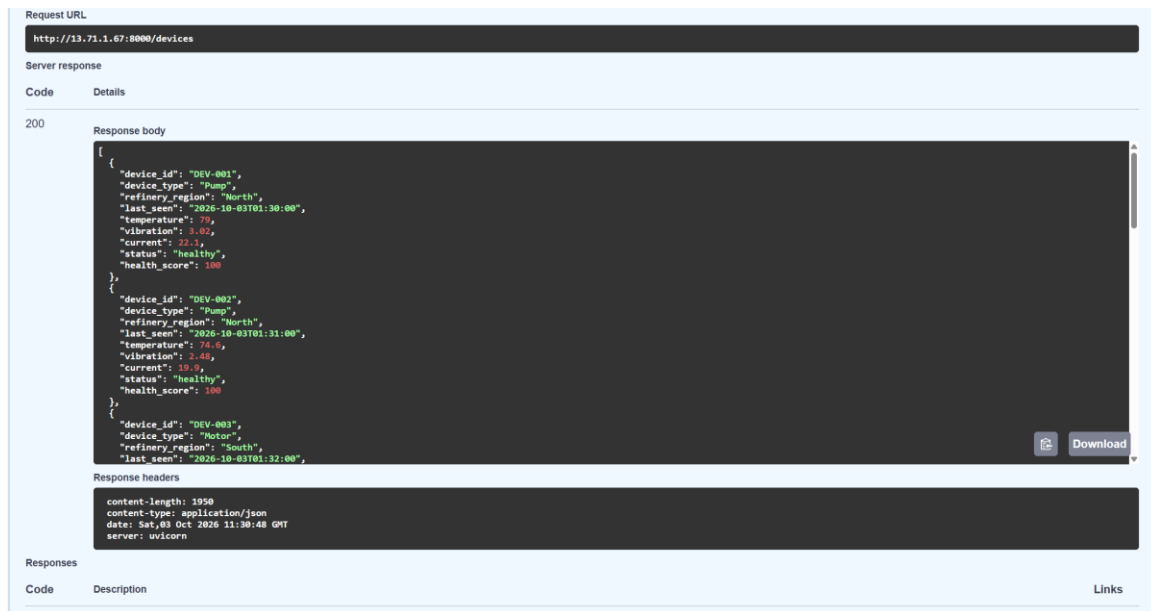
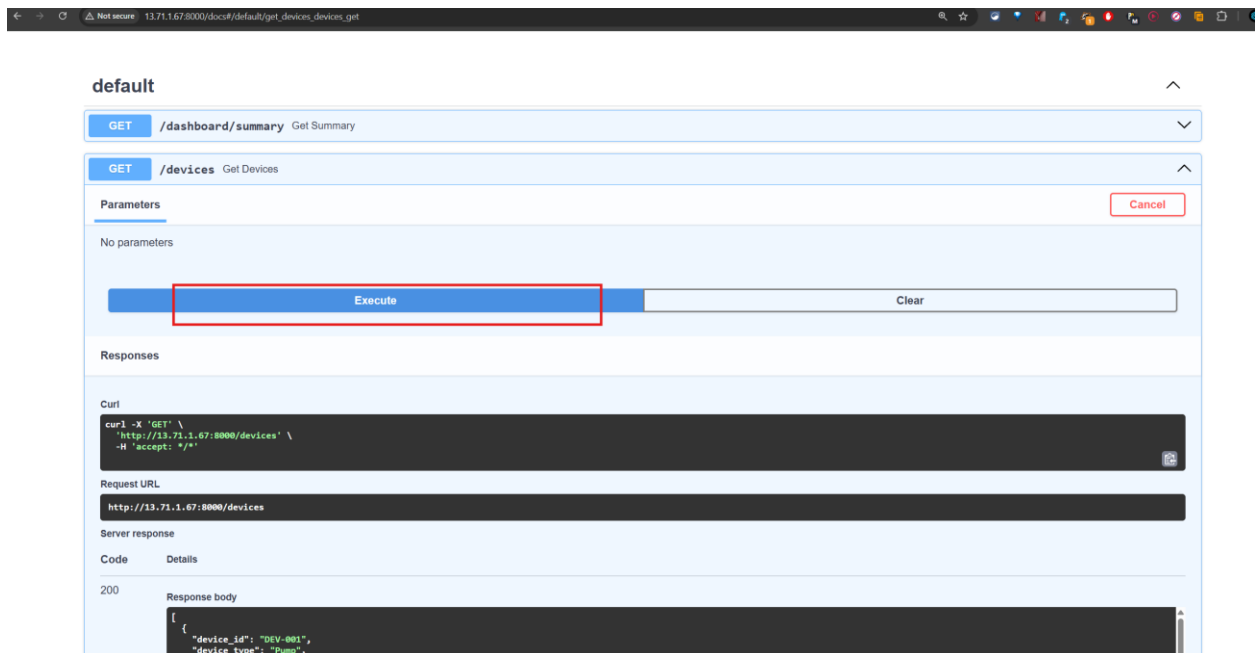
If you get below then it is working



Open Swagger using <http://13.71.1.67:8000/docs>



Execute



Once it shows the data that means backend is successfully connected to PostgreSQL Database.

FRONTEND

1. SSH into Frontend VM

```
ssh ssadmin@20.197.25.1
```

2. Clone the frontend

```
git clone https://github.com/devopsinsiders/axion-ui.git
```

```
cd axion-ui
```

3. Install Node.js 22

```
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -  
sudo apt install -y nodejs
```

Verify:

```
node -v
```

```
npm -v
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ node -v  
v22.23.3  
10.9.9  
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

4. Install dependencies

```
npm install (Do not run npm audit fix)
```

5. Configure Backend API

Do this before npm run build.

First check where the old API URL exists:

```
grep -Rni "api.axionsystems.de" src/
```

Then replace it:

```
sed -i 's|https://api.axionsystems.de|http://13.71.1.67:8000|g' \
```

```
src/App.tsx \
```

```
src/components/pages/DashboardView.tsx \
```

```
src/components/pages/HistoricalTrends.tsx
```

Verify:

```
grep -Rni "13.71.1.67" src/
```

You should see the new backend URL.

```
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ grep -Rni "api.axionsystems.de" src/
src/components/pages/DashboardView.tsx:10:const API_BASE = 'https://api.axionsystems.de';
src/components/pages/HistoricalTrends.tsx:7:const API_BASE = 'https://api.axionsystems.de';
src/App.tsx:18:const API_BASE = 'https://api.axionsystems.de';
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ sed -i 's|https://api.axionsystems.de|http://13.71.1.67:8000|g' \
src/App.tsx \
src/components/pages/DashboardView.tsx \
src/components/pages/HistoricalTrends.tsx
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ grep -Rni "13.71.1.67" src/
src/components/pages/DashboardView.tsx:10:const API_BASE = 'http://13.71.1.67:8000';
src/components/pages/HistoricalTrends.tsx:7:const API_BASE = 'http://13.71.1.67:8000';
src/App.tsx:18:const API_BASE = 'http://13.71.1.67:8000';
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

6. Build the frontend

npm run build

Verify the build:

```
ls -la dist/
```

```
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ ls -la dist/
total 144
drwxrwxr-x 3 ssadmin ssadmin 4096 Oct 3 11:42 .
drwxrwxr-x 8 ssadmin ssadmin 4096 Oct 3 11:42 ..
drwxrwxr-x 2 ssadmin ssadmin 4096 Oct 3 11:42 assets
-rw-rw-r-- 1 ssadmin ssadmin 7599 Oct 3 11:42 devopsinsiders-logo-light.png
-rw-rw-r-- 1 ssadmin ssadmin 284 Oct 3 11:42 favicon.svg
-rw-rw-r-- 1 ssadmin ssadmin 5031 Oct 3 11:42 icons.svg
-rw-rw-r-- 1 ssadmin ssadmin 551 Oct 3 11:42 index.html
-rw-rw-r-- 1 ssadmin ssadmin 109539 Oct 3 11:42 logo.png
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ ls
Dockerfile  dist          index.html  node_modules package.json  src          tsconfig.json  vite.config.ts
README.md   eslint.config.js nginx.conf  package-lock.json public         tsconfig.app.json tsconfig.node.json
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

And confirm the backend IP made it into the compiled files:

```
grep -Rni "13.71.1.67" dist/
```

```
grep -Rqs "13.71.1.67" dist/ && echo "IP FOUND" || echo "IP NOT FOUND"
```

7. Install Nginx

```
sudo apt update
```

```
sudo apt install -y nginx
```

Check:

```
sudo systemctl status nginx
```

```
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-10-03 11:44:59 UTC; 6s ago
     Docs: man:nginx(8)
   Process: 7147 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 7149 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 7179 (nginx)
    Tasks: 3 (limit: 9490)
   Memory: 2.4M (peak: 5.4M)
      CPU: 23ms
   CGroup: /system.slice/nginx.service
           └─7179 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─7182 "nginx: worker process"
               └─7183 "nginx: worker process"

Oct 03 11:44:59 frontend-vm-axion-dev-01 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 03 11:44:59 frontend-vm-axion-dev-01 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

8. Deploy the React/Vite build

You only need **one** copy command:

```
sudo rm -rf /var/www/html/*
```

```
sudo cp -r dist/* /var/www/html/
```

Then:

```
ls -la /var/www/html
```

```
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ ls -la /var/www/html
total 144
drwxr-xr-x 3 root root 4096 Oct 3 11:45 .
drwxr-xr-x 3 root root 4096 Oct 3 11:44 ..
drwxr-xr-x 2 root root 4096 Oct 3 11:45 assets
-rw-r--r-- 1 root root 7599 Oct 3 11:45 devopsinsiders-logo-light.png
-rw-r--r-- 1 root root 284 Oct 3 11:45 favicon.svg
-rw-r--r-- 1 root root 5031 Oct 3 11:45 icons.svg
-rw-r--r-- 1 root root 551 Oct 3 11:45 index.html
-rw-r--r-- 1 root root 109539 Oct 3 11:45 logo.png
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

9. Restart/reload Nginx

`sudo systemctl reload nginx`

If Nginx wasn't previously running:

`sudo systemctl enable --now nginx`

10. Test locally

`curl -I http://localhost`

Expected something similar to:

HTTP/1.1 200 OK

Server: nginx

```
ssadmin@frontend-vm-axion-dev-01:~/axion-ui$ curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Sat, 03 Oct 2026 11:46:22 GMT
Content-Type: text/html
Content-Length: 551
Last-Modified: Sat, 03 Oct 2026 11:45:41 GMT
Connection: keep-alive
ETag: "6ac0eae5-227"
Accept-Ranges: bytes

ssadmin@frontend-vm-axion-dev-01:~/axion-ui$
```

11. Allow HTTP port 80

On the **Frontend VM's Azure NSG**, allow:

Inbound

Protocol: TCP

Port: 80

Source: Internet

Action: Allow

Home > Resource Manager | Resource groups > rg-axion-dev-01 > frontend-vm-axion-dev-01

frontend-vm-axion-dev-01 | Network settings

Virtual machine

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Connect
Bastion
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings
Disks
Extensions + applications
Operating system
Configuration
Advisor recommendations
Properties
Locks

Attach network interface | Detach network interface | View topology | Troubleshoot | Refresh | Give feedback

Network interface / IP configuration
nic-frontend-vm-axion-dev-01 (primary) / internal (primary)

Essentials

Network interface	: nic-frontend-vm-axion-dev-01	Load balancers	: 0 (Configure)
Virtual network / subnet	: vnet-axion-dev-01 / frontend-subnet-axion-dev-01	Application security groups	: 0 (Configure)
Public IP address	: 20.197.25.1	Network security group	: nsg-frontend-axion-dev-01
Private IP address	: 10.0.1.4	Accelerated networking	: Disabled
Admin security rules	: 0 (Configure)	Effective security rules	: 0

Rules

Network security group nsg-frontend-axion-dev-01 (attached to networkInterface: nic-frontend-vm-axion-dev-01)
Impacts 0 subnets, 1 network interfaces

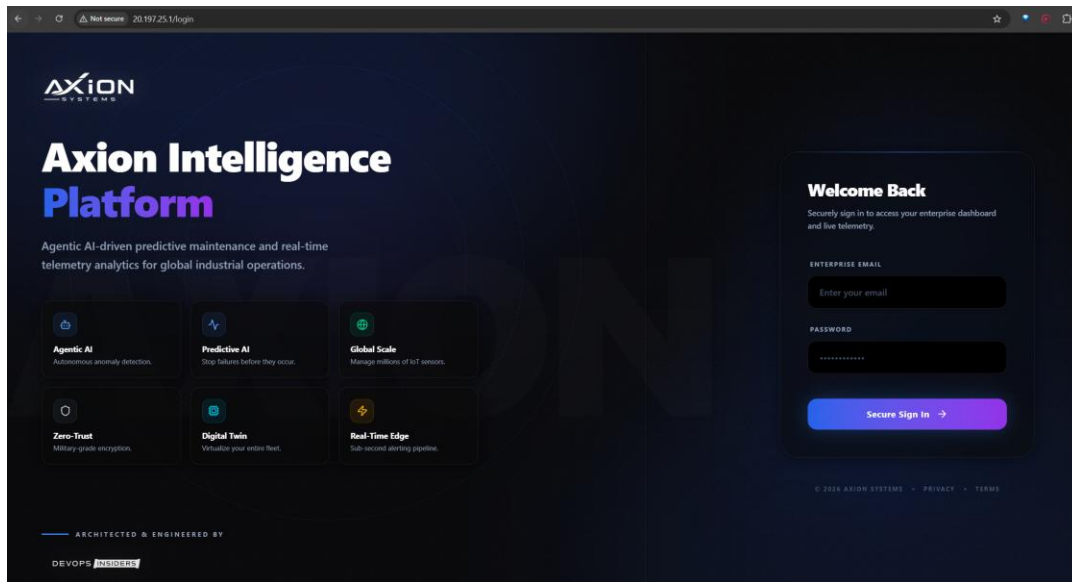
Search rules

Source == all | Destination == all | Protocol == all | Action == all | Port == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
100	AllowSSH	22	TCP	Any	Any	Allow
110	Allow80	80	Any	104.28.244.14	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

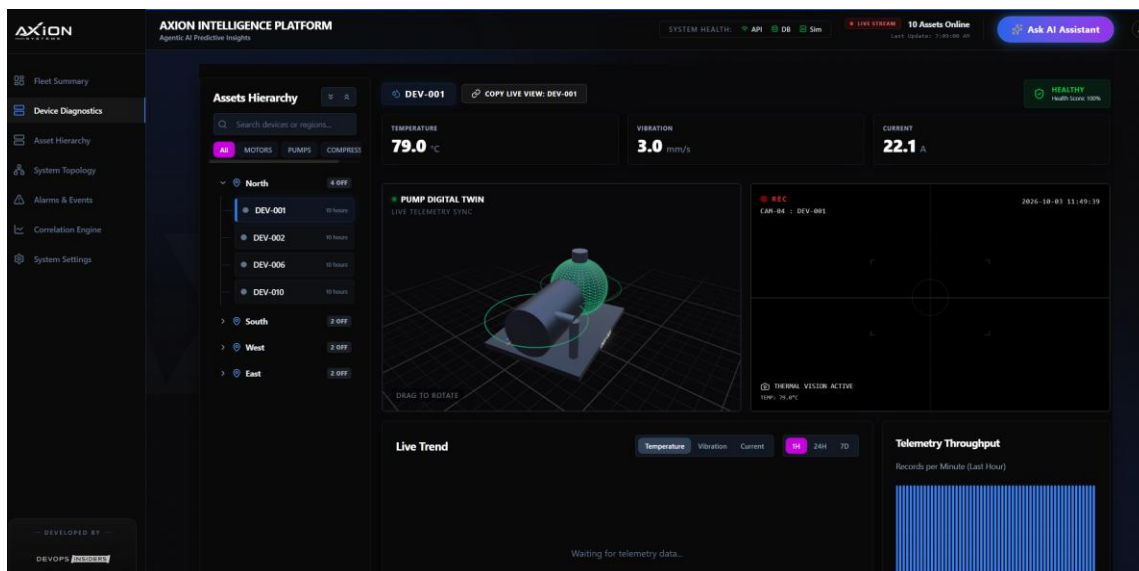
Then access:

http://<Frontend-Public-IP>



Email- info@devopsinsiders.com

P@ssw01rd@123



AXION

AXION INTELLIGENCE PLATFORM

Agents: AI Predictive Insights

SYSTEM HEALTH: API DB Sim LIVE STREAM 10 Assets Online

Ask AI Assistant

Fleet Summary

Device Diagnostics

Asset Hierarchy

System Topology

Alarms & Events

Correlation Engine

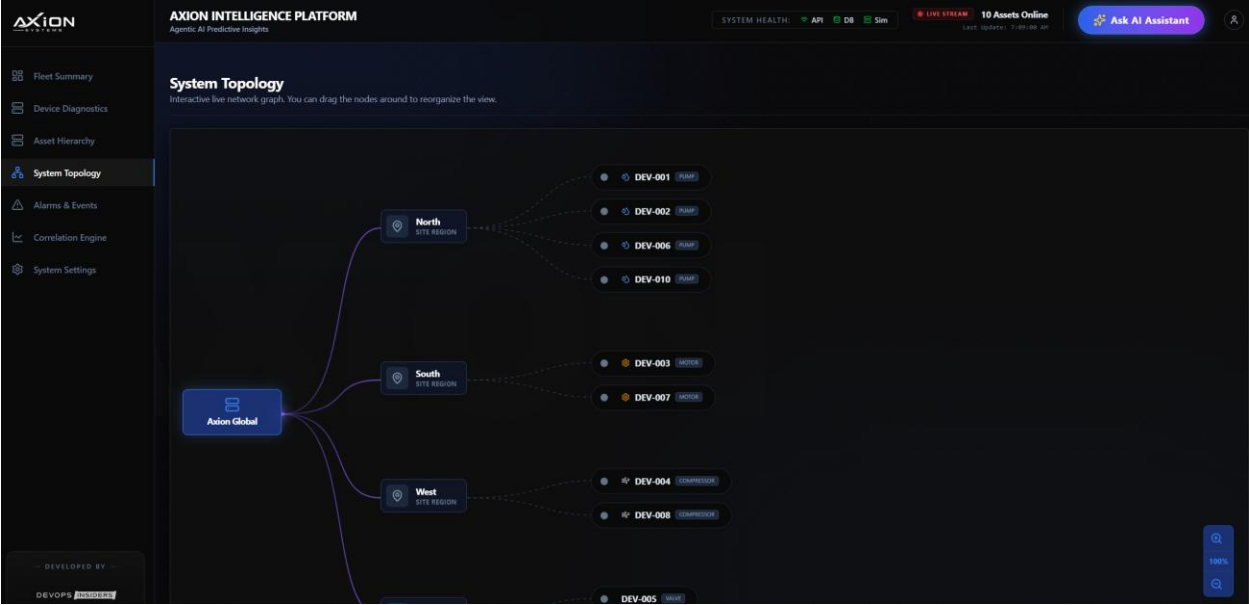
System Settings

Global Asset Hierarchy

Deep dive structural overview of all plant telemetry.

Search devices or regions...

ASSET / STRUCTURE	TYPE	STATUS	TEMP (°C)	VIB (MM/S)	CURRENT (A)
Enterprise: Axion Global Total Devices: 10					
North 4 Total					
DEV-001	PUMP	OFFLINE	-	-	-
DEV-002	PUMP	OFFLINE	-	-	-
DEV-006	PUMP	OFFLINE	-	-	-
DEV-010	PUMP	OFFLINE	-	-	-
South 2 Total					
DEV-003	MOTOR	OFFLINE	-	-	-
DEV-007	MOTOR	OFFLINE	-	-	-
West 2 Total					
DEV-004	COMPRESSOR	OFFLINE	-	-	-
DEV-008	COMPRESSOR	OFFLINE	-	-	-



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Alarms & Events

Historical log of system alerts and state changes.

ID	TIME	LEVEL	DEVICE	MESSAGE
ALH-001	7:07:00 AM	WARNING	DEV-008	Temperature warning (91.1°C)
ALH-002	7:02:00 AM	WARNING	DEV-003	Temperature warning (89.2°C)
ALH-003	7:06:00 AM	WARNING	DEV-007	Temperature warning (87.1°C)